Kaustubh Dua

Frontend Developer · Aspiring Software Engineer

kaustubhdua1991@gmail.com · github.com/kaustubhdua · LinkedIn

PROFILE

Electronics & Computer Engineering undergraduate building interactive web experiences and practical software across full-stack development, applied AI, data and systems. Interested in software engineering internships spanning frontend, backend, full-stack and applied AI/data. Values clear system boundaries, careful validation, accessible UX and tested releases.

EDUCATION

Jaypee Institute of Information Technology (JIIT), Noida — B.Tech., Electronics & Computer Engineering (ECM) · 2nd year · Expected 2029

EXPERIENCE

Pinnacle Labs · **Web Development Intern** · 2026
Built **WEATHER_CORE (Matrix Edition)**, a real-time weather dashboard with asynchronous Fetch API updates and a responsive HTML/CSS/JavaScript interface; and **Nord Studio Task Manager**, with localStorage persistence, priority labels and All/Active/Completed filters.

Selected work: Weather Core Matrix · Nord Studio Task Manager

SELECTED PROJECTS

NIRDHOOM · Multi-contributor prototype · React 19, Supabase, PostgreSQL/PostGIS, OR-Tools, PWA
Field-first crop-residue coordination across farmer bookings, operator dispatch, field evidence and residue-buyer pathways. Hosted prototype uses sample data; no impact metrics claimed. GitHub · Hosted prototype

Saathi · Multi-contributor MVP · Next.js, FastAPI, Python
Hindi/Hinglish assistance with deterministic intent routing, a 120-case synthetic benchmark, source provenance, consent prompts and human escalation. No public demo currently advertised. GitHub

Smart Waste · Engineering prototype · Python, scikit-learn, OR-Tools, Streamlit, Folium
Synthetic-data pipeline combining Random Forest fill-level prediction, collection-priority scoring and capacity-aware vehicle routing. No always-on hosted demo. GitHub

PatchPilot · Personal early-stage project · Python, FastAPI, pytest, Docker
Deterministic pull-request diff triage that emits evidence-backed risk signals and test suggestions with bounded inputs and rate limiting; not a security verdict. GitHub

GenomeGuard · Research prototype · Python, Starlette, NumPy, PostgreSQL
Genomic-data governance research with policy checks, heuristic disclosure-risk signals, output protection and a hash-chained audit trail on public 1000 Genomes data; not for clinical use. GitHub · Research instance

JIIT Youth Club Website · Multi-contributor club website · React 19, Supabase, Playwright, CodeQL
Public site for club information, events, gallery, archive and recruitment, supported by browser QA and security scanning. Current production deployment unverified. GitHub

SKILLS

Languages: Python, TypeScript, JavaScript, C++17, Java, SQL, HTML, CSS
Frontend: React, Next.js, Vite, Three.js/WebGL, Leaflet, Tailwind CSS, GSAP
Backend & Data: FastAPI, PostgreSQL/PostGIS, Supabase, SQLAlchemy, OR-Tools, scikit-learn, PyTorch, OpenCV, Ollama
Engineering: Git, GitHub Actions, pytest, Playwright, Appium, Docker, CodeQL

CERTIFICATES & CHALLENGES

- **JPMorgan Chase & Co. Software Engineering Job Simulation** (Forage, Apr–Jun 2026): Java/Spring Boot, Kafka, H2 and REST API integration.
- **Harvard CS50 Introduction to Databases with SQL:** schema design, joins, indexing and query optimisation.
- **OSDHACK'26:** Certificate of Merit for participation; team-built Adapt AI, a hardware-aware local Ollama assistant.
- **ISRO Bharatiya Antariksh Hackathon 2026:** participation with AkashChalak, a satellite-assisted air-quality and hotspot prototype.
- **Nexus AI Quiz Ignite 2026:** competition participant.

LEADERSHIP & COMMUNITY

JIIT Youth Club (JYC), Tech Vertical — Member, Tech and Management team · **AIML and AI-Tronics Club, JIIT** — Member, Tech and Management team · **National Service Scheme (NSS)** — Volunteer, Cinematography and Management